



EXAM PAPERS PRACTICE

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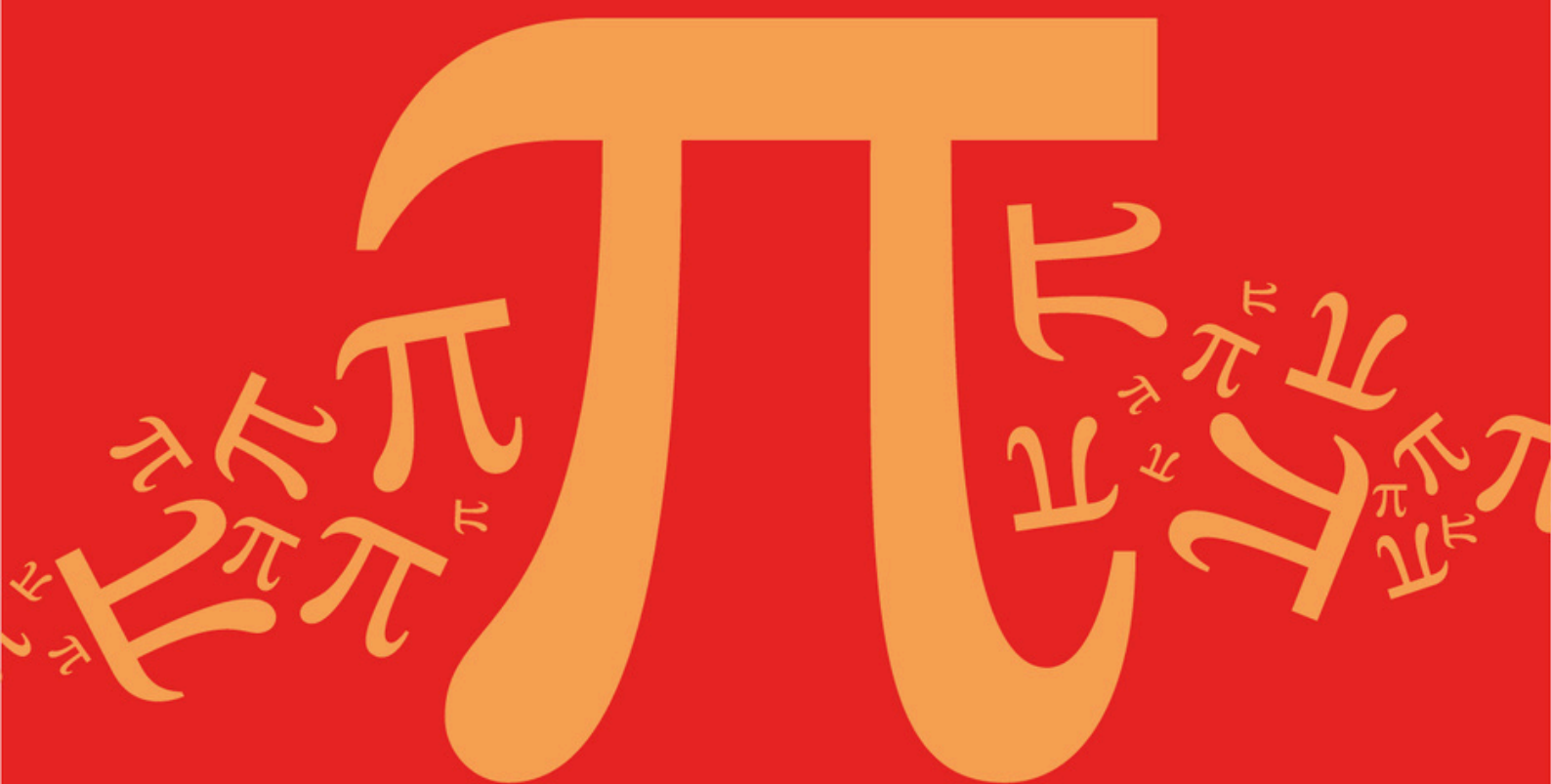
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D5: Find Maclaurin Series

Mark Scheme

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Topic	D: Further Algebra and Functions
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Ignore higher powers beyond x^2 throughout this question

(a) $e^{3x} = 1 + 3x + 4.5x^2$

B1

1

(b) $(1 + 2x)^{-3/2} = 1 - 3x + \frac{15}{2}x^2$

$(1 + 2x)^{-3/2} = 1 \pm 3x + kx^2$ or $1 + kx \pm 7.5x^2$ OE

M1

$1 - 3x + 7.5x^2$ OE (simplified PI)

A1

$e^{3x} (1 + 2x)^{-3/2} = (1 + 3x + 4.5x^2)(1 - 3x + 7.5x^2)$

Product of c's two expansions with an attempt to multiply out to find x^2 term

M1

x^2 term(s): $7.5x^2 - 9x^2 + 4.5x^2 = 3x^2$.

A1

4

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[5]

Q2.

(a) (i) $y = \ln(e^{3x} \cos x) = \ln e^{3x} + \ln \cos x = 3x + \ln \cos x$

B1

$\frac{dy}{dx} = 3 + \frac{1}{\cos x} \times (-\sin x)$

Chain rule for derivative of $\ln \cos x$

M1

$\frac{dy}{dx} = 3 - \tan x$

CSO AG

A1

3

(ii) $\frac{d^2y}{dx^2} = -\sec^2 x$; $\frac{d^3y}{dx^3} = -2\sec x(\sec x \tan x)$

$$M1 \text{ for } d/dx \{ [f(x)]^2 \} = 2f(x)f'(x)$$

B1; M1

$$\frac{d^4 y}{dx^4} = -4 \sec x (\sec x \tan x) \tan x - 2 \sec^4 x$$

ACF

A1

3

(b) Maclaurin's Thm:

$$y(0) + x y'(0) + \frac{x^2}{2!} y''(0) + \frac{x^3}{3!} y'''(0) + \frac{x^4}{4!} y^{(iv)}(0)$$

$$y(0) = \ln 1 = 0; y'(0) = 3; y''(0) = -1; y'''(0) = 0; y^{(iv)}(0) = -2$$

Mac. Thm with attempt to evaluate at least two derivatives at $x = 0$

M1

$$\ln(e^{3x} \cos x) = 0 + 3x + \frac{-1}{2!} x^2 + \frac{0}{3!} x^3 + \frac{-2}{4!} x^4 \dots$$

At least 3 of 5 terms correctly obtained.
Ft one miscopy in (a)

A1F

$$= 3x - \frac{1}{2} x^2 - \frac{1}{12} x^4$$

CSO AG Be convinced

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A1

3

$$(c) \{ \ln(1+px) \} = px - \frac{1}{2} p^2 x^2$$

accept $(px)^2$ for $p^2 x^2$; ignore higher powers;

B1

1

$$(d) (i) \left[\frac{1}{x^2} \{ \ln(e^{3x} \cos x) - \ln(1+px) \} \right] =$$

$$\left[\frac{1}{x^2} \left\{ 3x - \frac{1}{2} x^2 - O(x^4) - \left(px - \frac{1}{2} p^2 x^2 + O(x^3) \right) \right\} \right]$$

Law of logs and expansions used;

M1

$$\lim_{x \rightarrow 0} \left[\frac{1}{x^2} \ln \left(\frac{e^{3x} \cos x}{1+px} \right) \right] \text{ to exist, } p = 3$$

$p = 3$ convincingly found

A1

$$(ii) \dots\dots = \lim_{x \rightarrow 0} \left[\left(\frac{3-p}{x} \right) - \frac{1}{2} + \frac{p^2}{2} - O(x) \right]$$

Divide throughout by x^2 before taking limit. (m1 can be awarded before or after the A1 above)

m1

$$\text{Value of limit} = -\frac{1}{2} + \frac{p^2}{2} = 4$$

Must be convincingly obtained

A1

4

[14]

Q3.

$$(a) \quad y = (4 + \sin x)^{1/2} \text{ so } y^2 = 4 + \sin x$$

$$2y \frac{dy}{dx} = \cos x$$

$$\frac{dy}{dx}(y)^2 = 2y \frac{dy}{dx}$$

M1

$$y \frac{dy}{dx} = \frac{1}{2} \cos x$$

A1

2

Alternative:

$$\frac{dy}{dx} = \frac{1}{2} (4 + \sin x)^{-1/2} (\cos x)$$

chain rule

(M1)

$$y \frac{dy}{dx} = \frac{1}{2} \cos x$$

(A1)

(2)

$$(b) \quad y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = -\frac{1}{2} \sin x$$

Correct differentiation of $y \frac{dy}{dx}$

M1

When $x = 0, y = 2, \frac{dy}{dx} = \frac{1}{4}, 2 \frac{d^2y}{dx^2} + \left(\frac{1}{4}\right)^2 = 0$
Ft on RHS of M1 line as $k \sin x$

A1F

$$y \frac{d^3y}{dx^3} + \frac{dy}{dx} \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} \frac{d^2y}{dx^2} = -\frac{1}{2} \cos x$$

Correct LHS

m1

A1

When $x = 0, 2 \frac{d^3y}{dx^3} + 3 \left(\frac{1}{4}\right) \left(-\frac{1}{32}\right) = -\frac{1}{2} \Rightarrow \frac{d^3y}{dx^3} = -\frac{61}{256}$
CSO

A1

5

Alternative:

$$\frac{d^2y}{dx^2} = -\frac{1}{4} (4 + \sin x)^{-3/2} (\cos^2 x) + \frac{1}{2} (4 + \sin x)^{-1/2} (-\sin x)$$

Sign and numerical coeffs errors only.

(M1)

ACF

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(A1)

$$\frac{d^3y}{dx^3} = \frac{3}{8} (4 + \sin x)^{-2.5} (\cos^3 x) - \frac{1}{4} (4 + \sin x)^{-1.5} (-2 \cos x \sin x)$$

Sign and numerical coeffs errors only.

(m1)

$$- \frac{1}{4} (4 + \sin x)^{-1.5} (\cos x) (-\sin x) - \frac{1}{2} (4 + \sin x)^{-0.5} \cos x$$

ACF

(A1)

When $x = 0, \frac{d^3y}{dx^3} = \frac{3}{8} \times \frac{1}{32} - \frac{1}{2} \times \left(\frac{1}{2}\right) = -\frac{61}{256}$
CSO

(A1)

(5)

(c) McC. Thm: $y(0) + x y'(0) + \frac{x^2}{2} y''(0) + \frac{x^3}{3!} y'''(0)$

Maclaurin's theorem used with c's numerical values
for $y(0)$, $y'(0)$, $y''(0)$ and $y'''(0)$ all found with at least three
being non-zero.

M1

$$(4 + \sin x)^{1/2} \approx 2 + \frac{1}{4}x - \frac{1}{64}x^2 - \frac{61}{1536}x^3 \dots$$

CSO Previous 6 marks must have been scored

A1

2

[9]

Q4.

(a) $y = \ln \cos 2x \Rightarrow y'(x) = \frac{1}{\cos 2x} (-2 \sin 2x)$
Chain rule

M1

A1

$$y''(x) = -4 \sec^2 2x$$

$\lambda \sec^2 2x$ OE

m1

$$y'''(x) = -8 \sec 2x (2 \sec 2x \tan 2x)$$

$K \sec^2 2x \tan 2x$ OE

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M1

$$\{y''''(x) = -16 \tan 2x (\sec^2 2x)\}$$

$$y''''(x) = -16[2 \sec^2 2x (\sec^2 2x) + \tan 2x (2 \sec 2x (2 \sec 2x \tan 2x))]$$

Product rule OE

M1

ACF

A1

6

(b) $y(0) = 0, y'(0) = 0, y''(0) = -4, y'''(0) = 0, y''''(0) = -32$
ft c's derivatives

B1F

$$\ln \cos 2x \approx 0 + 0 + \frac{x^2}{2}(-4) + 0 + \frac{x^4}{4!}(-32)$$

M1

$$\approx -2x^2 - \frac{4}{3}x^4$$

CSO throughout parts (a) and (b) AG

A1

3

(c) $\ln(\sec^2 2x) = -2\ln(\cos 2x)$

PI

M1

$$\approx 4x^2 + \frac{8}{3}x^4$$

A1

2

[11]

Q5.

(a) $\sin 2x = 2x - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!}$

Accept ACF even if unsimplified

$$= 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$$

B1

1

(b) $\lim_{x \rightarrow 0} \left[\frac{2x - \sin 2x}{x^3 \ln(1+kx)} \right] = \lim_{x \rightarrow 0} \frac{2x - (2x - \frac{4}{3}x^3 + \frac{4}{15}x^5 \dots)}{x^2 \left(kx - \frac{(kx)^2}{2} + \dots \right)}$

Using series expansions.

M1

Expansion of $\ln(1+kx) = kx (-\dots)$

B1

$$= \lim_{x \rightarrow 0} \left[\frac{\frac{4}{3}x^3 - \frac{4}{15}x^5 + \dots}{kx^3 - \frac{k^2}{2}x^4} \right] = \lim_{x \rightarrow 0} \left[\frac{\frac{4}{3} - O(x^2)}{k - O(x)} \right]$$

Dividing numerator and denominator by x^3 to get constant term in each. Must be at least a total of 3 terms divided by x^3

m1

$$\frac{4}{3k} = 16 \Rightarrow k = \frac{1}{12}$$

OE exact value. Dep on numerator being of form $4 / 3(OE) + \lambda x^2 \dots (\lambda \neq 0)$ and denominator being of form $k + \mu x \dots (\mu \neq 0)$ before limit taken

A1

4

[5]

Q6.

(a) $y = \ln(1 + \sin x), \quad \frac{dy}{dx} = \frac{1}{1 + \sin x} \times (\cos x)$

Chain rule OE

ACF eg $e^{-y} \cos x$

M1

A1

2

(b) $\left(\frac{d^2 y}{dx^2} \right) = \frac{(1 + \sin x)(-\sin x) - \cos x(\cos x)}{(1 + \sin x)^2}$

Quotient rule OE, with u and v non constant

M1

ACF

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A1

$$\frac{d^2 y}{dx^2} = \frac{-\sin x - 1}{(1 + \sin x)^2} = \frac{-1}{1 + \sin x} = \frac{-1}{e^y} = e^{-y}$$

CSO AG Completion must be convincing

A1

3

(c) $\frac{d^3 y}{dx^3} = e^{-y} \frac{dy}{dx}$

ACF for $\frac{d^3 y}{dx^3}$

B1

$$\frac{d^4 y}{dx^4} = -e^{-y} \left(\frac{dy}{dx} \right)^2 + e^{-y} \frac{d^2 y}{dx^2}$$

Product rule OE and chain rule

M1

$$\frac{d^4 y}{dx^4} = -e^{-y} \left(\frac{dy}{dx} \right)^2 - (e^{-y})^2$$

OE in terms of e^{-y} and $\frac{dy}{dx}$ only

A1

3

(d) $y(0) = 0; \quad y'(0) = 1; \quad y''(0) = -1;$

Ft only for $y'(0)$; other two values must be correct

B1F

$$y(x) \approx$$

$$y(0) + xy'(0) + \frac{x^2}{2} y''(0) + \frac{x^3}{3!} y'''(0) + \frac{x^4}{4!} y^{(iv)}(0)$$

$$y'''(0) = 1; \quad y^{(iv)}(0) = -2$$

Maclaurin's theorem applied with numerical values for $y'(0), y''(0), y'''(0)$, and $y^{(iv)}(0)$. M0 if missing an expression for any one of the 1st, 3rd or 4th derivatives

M1

$$\ln(1 + \sin x) \approx x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{12}x^4 \dots$$

A0 if FIW

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3

[11]

Q7.

(a) (i) $\cos x + \sin x = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3$

Accept coeffs unsimplified, even 3! for 6.

B1

1

(ii) $\ln(1 + 3x) = 3x - \frac{1}{2}(3x)^2 + \frac{1}{3}(3x)^3 = 3x - \frac{9}{2}x^2 + 9x^3$

Accept coeffs unsimplified

B1

1

(b) (i) $y = e^{\tan x}, \quad \frac{dy}{dx} = \sec^2 x e^{\tan x}$

Chain rule

M1

ACF eg $y \sec^2 x$

A1

$$\frac{d^2 y}{dx^2} = 2 \sec^2 x \tan x e^{\tan x} + \sec^4 x e^{\tan x}$$

Product rule OE

m1

ACF

A1

$$= \sec^2 x e^{\tan x} (2 \tan x + \sec^2 x)$$

$$= \frac{dy}{dx} (2 \tan x + 1 + \tan^2 x)$$

$$\frac{d^2 y}{dx^2} = (1 + \tan x)^2 \frac{dy}{dx}$$

AG Completion; CSO any valid method.

A1

5

$$(ii) \quad \frac{d^3 y}{dx^3} = 2(1 + \tan x) \sec^2 x \frac{dy}{dx} + (1 + \tan x)^2 \frac{d^2 y}{dx^2}$$

M1

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$$\text{When } x = 0, \quad \frac{d^3 y}{dx^3} = 2(1)(1)(1) + (1)(1) = 3$$

CSO

A1

2

$$(iii) \quad y(0) = 1; \quad y'(0) = 1; \quad y''(0) = 1; \quad y'''(0) = 3;$$

$$y(x) \approx y(0) + xy'(0) + \frac{1}{2} x^2 y''(0) + \frac{1}{3!} x^3 y'''(0)$$

M1

$$e^{\tan x} \approx 1 + x + \frac{1}{2} x^2 + \frac{1}{2} x^3$$

CSO AG

A1

2

(c) $\lim_{x \rightarrow 0} \left[\frac{e^{\tan x} - (\cos x + \sin x)}{x \ln(1+3x)} \right]$

$$= \lim_{x \rightarrow 0} \frac{1+x+\frac{x^2}{2}+\frac{x^3}{2}-1-x+\frac{x^2}{2}+\frac{x^3}{6}}{x\left(3x-\frac{9}{2}x^2+\dots\right)}$$

Using series expns.

M1

$$= \lim_{x \rightarrow 0} \left[\frac{x^2 + \frac{2}{3}x^3 + \dots}{3x^2 - \frac{9}{2}x^3 \dots} \right] = \lim_{x \rightarrow 0} \left[\frac{1 + \frac{2}{3}x + \dots}{3 - \frac{9}{2}x \dots} \right]$$

Dividing numerator and denominator by x^2 to get constant terms. OE following a slip.

m1

$$= \frac{1}{3}$$

A1

3

EXAM PAPERS PRACTICE [14]

Q8.

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(a) $\frac{dy}{dx} = \frac{2 \sec^2 x}{1+2 \tan x}$

Chain rule

M1

ACF for $y'(x)$

A1

$$\frac{d^2 y}{dx^2} = \frac{(1+2 \tan x)(4 \sec^2 x \tan x) - 2 \sec^2 x(2 \sec^2 x)}{(1+2 \tan x)^2}$$

Quotient rule OE in which both u and v are not const. or applied to a correct form of y'

ACF for $y''(x)$

M1

A1

4

(b) McC. Thm: $y(0) + x y'(0) + \frac{x^2}{2} y''(0)$

$(y(0) = 0); \quad y'(0) = 2; \quad y''(0) = -4$

Attempt to evaluate at least $y'(0)$ and $y''(0)$. PI

M1

$\ln(1 + 2 \tan x) \approx 2x - 2x^2$

Dep on previous 5 marks

A1

2

(c) $\ln(1-x) = -x - \frac{1}{2}x^2 \dots$

Ignore higher power terms

B1

$$\left[\frac{\ln(1+2\tan x)}{\ln(1-x)} \right] \approx \frac{2x - 2x^2 \dots}{-x - \frac{1}{2}x^2 \dots}$$

Expansions used

M1

$$= \frac{2 - 2x \dots}{-1 - \frac{1}{2}x \dots}$$

Dividing num. and den. by x to get constant term in each and non-const term in at least num. or den.

m1

So $\lim_{x \rightarrow 0} \left[\frac{\ln(1+2\tan x)}{\ln(1-x)} \right] = \frac{2}{-1} = -2$

ft c's answer to (b) provided answer (b) is in the form $\pm px \pm qx^2$ and B1 awarded

A1F

4

[10]

Q9.

(a) $\frac{dy}{dx} = \frac{1}{4+3x} \times 3$

Chain rule

M1

$$\frac{d^2y}{dx^2} = -3(4+3x)^{-2} \times 3 = -9(4+3x)^{-2}$$

M1 for quotient (PI) or chain rule used

M1A1

3

(b) $\ln(4+3x) = \ln 4 + y'(0)x + y''(0)\frac{1}{2}x^2 + \dots$

*Clear attempt to use Maclaurin's theorem
with numerical values for $y'(0)$ and $y''(0)$*

M1

First three terms: $\ln\left(4 + \frac{3}{4}x - \frac{9}{32}x^2\right)$

*ft on c's answers to (a) provided $y'(0)$ and
 $y''(0)$ are $\neq 0$. Accept 1.38(6..) for $\ln 4$*

A1F

2

(c) $\ln(4-3x) = \ln 4 - \frac{3}{4}x - \frac{9}{32}x^2$

ft $x \rightarrow -x$ in c's answer to (b)

B1F

1

(d) $\ln\left(\frac{4+3x}{4-3x}\right) = \ln(4+3x) - \ln(4-3x)$

M1

$$\approx \ln 4 + \frac{3}{4}x - \frac{9}{32}x^2 - \ln 4 + \frac{3}{4}x + \frac{9}{32}x^2$$

$$\approx \frac{3}{2}x$$

CSO AG

A1

2

[8]

Q10.

(a) $\sin 3x = 3x - \frac{1}{3!}(3x)^3 + \dots$

B1

1

(b) $\cos 2x = 1 - \frac{1}{2!} (2x)^2 + \dots$

B1

$$\lim_{x \rightarrow 0} \left[\frac{3x \cos 2x - \sin 3x}{5x^3} \right] =$$

$$\lim_{x \rightarrow 0} \frac{3x - 6x^3 - 3x + 4.5x^3 + \dots}{5x^3}$$

Using expansions

M1

$$= \lim_{x \rightarrow 0} \frac{-1.5 + (0(x^2)) \dots}{5}$$

Division by x^3 stage to reach relevant form of quotient before taking limit.

m1

$$= -\frac{3}{10}$$

CSO OE

A1

EXAM PAPERS PRACTICE ⁴ [5]

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Q11.

(a) $\cos 4x \approx 1 - \frac{(4x)^2}{2} + \frac{(4x)^4}{4!} \dots$

Clear attempt to replace x by $4x$ in expansion of $\cos x$...condone missing brackets for the M mark

M1

$$\approx 1 - 8x^2 + \frac{32}{3}x^4 \dots$$

A1

2

(b) (i) $\frac{dy}{dx} = \frac{1}{2 - e^x} \times (-e^x)$

Chain rule

M1A1

$$\frac{d^2 y}{dx^2} = \frac{(2 - e^x)(-e^x) - (-e^x)(-e^x)}{(2 - e^x)^2}$$

Quotient rule OE

M1

ACF

A1

$$= \frac{-2e^x}{(2 - e^x)^2}$$

$$\frac{d^3 y}{dx^3} = \frac{(2 - e^x)^2(-2e^x) - (-2e^x)2(2 - e^x)(-e^x)}{(2 - e^x)^4}$$

All necessary rules attempted
(dep on previous 2 M marks)

m1

ACF

A1

6

(ii) $y(0) = 0; y'(0) = -1; y''(0) = -2; y'''(0) = -6$

At least three attempted

M1

$$\ln(2 - e^x) \approx y(0) + xy'(0) + \frac{x^2}{2} y''(0) + \dots$$

$$\dots \approx -x - x^2 - x^3 \dots$$

CSO AG (The previous 7 marks
must have been awarded and no
double errors seen)

A1

2

(c) $\left[\frac{x \ln(2 - e^x)}{1 - \cos 4x} \right]_{\infty} \frac{-x^2 - x^3 - x^4}{8x^2 - \frac{32}{3}x^4}$

Using the expansions

M1

$$\text{Limit} = \lim_{x \rightarrow 0} \frac{-x^2 - o(x^3)}{8x^2 - o(x^4)}$$

The notation $o(x^n)$ can be
replaced by a term of the form kx^n

$$\dots = \lim_{x \rightarrow 0} \frac{-1 - o(x)}{8 - o(x^2)}$$

Division by x^2 stage before taking the limit

m1

$$\dots = -\frac{1}{8}$$

CSO

A1

3

[13]

Q12.

(a) (i)

$$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$$

*Clear use of $x \rightarrow 2x$ in
expansion of e^x*

ACF

M1

A1

2

(ii) $\{f(x)\} = e^{2x}(1+3x)^{-\frac{2}{3}}$

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$$(1+3x)^{-\frac{2}{3}} = 1 + \left(-\frac{2}{3}\right)(3x) + \frac{\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)(3x)^2}{2} - \frac{40}{3}x^3$$

First three terms as

$$1 + \left(-\frac{2}{3}\right)(3x) + kx^2 \quad \text{OE}$$

M1

$$= 1 - 2x + 5x^2 - \frac{40}{3}x^3$$

A1

$$\{f(x)\} \approx$$

$$1 + 2x + 2x^2 + \frac{4x^3}{3} - 2x - 4x^2 - 4x^3 + 5x^2 + 10x^3 - \frac{40x^3}{3}$$

Dep on both prev MS

m1

Condone one sign or
numerical slip in mult.

A1Ft

$$= 1 + 3x^2 - 6x^3$$

CSO AG A0 if
binominal series not used

A1

5

(b) (i) $y = \ln(1 + 2 \sin x) \Rightarrow \frac{dy}{dx} = \frac{1}{1 + 2 \sin x} - 2 \cos x$

Chain rule

M1A1

$$\frac{d^2y}{dx^2} = \frac{(1 + 2 \sin x)(-2 \sin x) - 2 \cos x(2 \cos x)}{(1 + 2 \sin x)^2} = \frac{-2(\sin x + 2)}{(1 + 2 \sin x)^2}$$

Quotient rule OE with
u and v non constant

M1

ACF

A1

4

(ii) $y(0) = 0, \quad y'(0) = 2, \quad y''(0) = -4$

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M1

McL Thm: $\{\ln(1 + 2 \sin x)\} \approx 0 + 2x - 4\left(\frac{x^2}{2}\right) \dots \approx 2x - 2x^2$

CSO AG

A1

2

(c) $\lim_{x \rightarrow 0} \frac{1 - f(x)}{x \ln(1 + 2 \sin x)} = \lim_{x \rightarrow 0} \frac{-3x^2 + 6x^3}{2x^2 - 2x^3}$

Using expansions

M1

$$= \lim_{x \rightarrow 0} \frac{-3 + 6x}{2 - 2x}$$

Division by x^2 stage
before taking limit.

m1

$$= -\frac{3}{2}$$

CSO

A1

3

[16]

Q13.

(a) (i) $f(x) = (9 + \tan x)^{\frac{1}{2}}$
 so $f'(x) = \frac{1}{2}(9 + \tan x)^{-\frac{1}{2}} \sec^2 x$

Chain rule

M1A1

$$f''(x) = -\frac{1}{4}(9 + \tan x)^{-\frac{3}{2}} \sec^4 x + \frac{1}{2}(9 + \tan x)^{-\frac{1}{2}} (2 \sec^2 x \tan x)$$

Product rule, OE

M1

ACF

A1

4

(ii) $f(0) = 3$

B1

$$f'(0) = \frac{1}{2}(9)^{-\frac{1}{2}} = \frac{1}{6};$$

Both attempted and at least one correct ft
 on c's $f'(x)$ and $f''(x)$

M1

$$f''(0) = -\frac{1}{4}(9)^{-\frac{3}{2}} = -\frac{1}{108}$$

$$f(x) \approx f(0) + x f'(0) + \frac{1}{2} x^2 f''(0)$$

$$(9 + \tan x)^{\frac{1}{2}} \approx 3 + \frac{x}{6} - \frac{x^2}{216}$$

CSO AG

A1

3

$$(b) \quad \frac{f(x) - 3}{\sin 3x} \approx \frac{\frac{x}{6} - \frac{x^2}{216} \dots}{3x - \frac{(3x)^3}{3!} \dots}$$

Using series expns.

M1

$$\approx \frac{\frac{1}{6} - \frac{x}{216} \dots}{3 - \dots}$$

Dividing numerator and denominator by x
to get constant term in each

m1

$$\lim_{x \rightarrow 0} \left[\frac{f(x) - 3}{\sin 3x} \right] = \frac{1}{18}$$

A1

3

[10]

Q14.

$$(a) \quad (i) \quad \ln(1 + 2x) = 2x - 2x^2 + \frac{8}{3}x^3 \dots$$

Use of expansion of $\ln(1 + x)$

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Simplified 'numerators'.

M1

A1

2

$$(ii) \quad -\frac{1}{2} < x \leq \frac{1}{2}$$

B1

1

$$(b) \quad (i) \quad y = \ln \cos x \Rightarrow y'(x) = \frac{1}{\cos x} (-\sin x)$$

M1

$$y''(x) = -\sec^2 x$$

ACF

A1

$$y''(x) = -2 \sec x (\sec x \tan x)$$

Chain rule OE

M1

$$\{y'''(x) = -2\tan x(\sec^2 x)\}$$

Ft a slip...accept unsimplified

A1ft

4

(ii) $y'''(x) = -2[\sec^2 x(\sec^2 x) + \tan x(2 \sec x(\sec x \tan x))]$

Product rule OE

M1

ACF

A1

$$y'''(0) = -2[(1)^2 + 0] = -2$$

Ft a slip

A1ft

3

(iii) $\ln \cos x \approx 0 + 0 + \frac{x^2}{2}(-1) + 0 \frac{x^4}{4!}(-2)$

M1

$$\approx -\frac{x^2}{2} - \frac{x^4}{12}$$

CSO throughout part (b). AG

A1

2

(c)
$$\text{Limit} = \lim_{x \rightarrow 0} \left[\frac{x \ln(1+2x)}{x^2 - \ln \cos x} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{x(2x - 2x^2 + \dots)}{x^2 - \left(-\frac{x^2}{2} - \frac{x^4}{12} \dots \right)} \right]$$

Using earlier expansions

M1

$$\text{Limit} = \lim_{x \rightarrow 0} \frac{2x^2 - o(x^3)}{1.5x^2 + o(x^4)}$$

The notation $o(x^n)$ can be replaced by a term of the form kx^n

$$= \lim_{x \rightarrow 0} \frac{2 - o(x)}{1.5 + o(x^2)} = \frac{4}{3}$$

Need to see stage, division by x^2

M1A1

3

[15]

Q15.

(a) $\sin 2x \approx 2x - \frac{(2x)^3}{3!} + \dots = 2x - \frac{4}{3}x^3 + \dots$

B1

1

(b) (i) $\frac{dy}{dx} = \frac{1}{2}(3 + e^x)^{-\frac{1}{2}}(e^x)$

Chain rule

M1A1

$$\frac{d^2y}{dx^2} = \frac{1}{2}e^x(3 + e^x)^{-\frac{1}{2}} - \frac{1}{4}(3 + e^x)^{-\frac{3}{2}}(e^{2x})$$

Product rule OE

M1

OE

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A1

$$y'(0) = \frac{1}{4}; \quad y''(0) = \frac{1}{4} - \frac{1}{32} = \frac{7}{32}$$

CSO

A1

5

(ii) $y(0) = 2; y'(0) = \frac{1}{4}; y''(0) = \frac{1}{4} - \frac{1}{32} = \frac{7}{32}$

$$\text{McC. Thm: } y(0) + xy'(0) + \frac{x^2}{2}y''(0)$$

$$\sqrt{3 + e^x} \approx 2 + \frac{1}{4}x + \frac{7}{64}x^2$$

CSO; AG

M1A1

2

(c)
$$\left[\frac{\sqrt{3+e^x} - 2}{\sin 2x} \right] = \left[\frac{2 + \frac{1}{4}x + \frac{7}{64}x^2 - 2}{2x - \frac{4}{3}x^3} \right]$$

M1

$$= \left[\frac{\frac{1}{4} + \frac{7}{64}x + \dots}{2 - \frac{4}{3}x^2 + \dots} \right]$$

Dividing numerator and denominator by x to get constant term in each

$$\lim_{x \rightarrow 0} \left[\frac{\sqrt{3+e^x} - 2}{\sin 2x} \right] = \frac{\frac{1}{4}}{\frac{2}{2}} = \frac{1}{8}$$

m1

Ft on cand's answer to (a) provided of the form $ax + bx^3$

A1F

3

[11]

Q16.

(a) (i) $f'(x) = \frac{1}{2}(1+2x)^{-\frac{1}{2}}(2) = (1+2x)^{-\frac{1}{2}}$

M1A1

$$f''(x) = -(1+2x)^{-\frac{3}{2}}$$

ft a slip

A1F

$$f'''(x) = 3(1+2x)^{-\frac{5}{2}}$$

A1

4

(ii) $f(x) = (1+2x)^{\frac{1}{2}} \Rightarrow f(0) = 1;$

B1

$$f'(0) = 1; f''(0) = -1; f'''(0) = 3$$

All three attempted

M1

ft on $k(1 + 2x)^m$

A1F

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2} f''(0) + \frac{x^3}{6} f'''(0)$$

$$\dots \approx 1 + x - \frac{x^2}{2} + \frac{x^3}{2}$$

CSO **AG**

A1

4

(b) $e^x (1 + 2x)^{\frac{1}{2}} \approx$

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right) \left(1 + x - \frac{x^2}{2} + \frac{x^3}{2}\right)$$

Attempt to expand needed

M1

$$\approx 1 + x(1 + 1) + x^2(-0.5 + 1 + 0.5)$$

A1

$$+ x^3 \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{2} + \frac{1}{6} \right)$$

$$\approx 1 + 2x + x^2 + \frac{2}{3}x^3$$

CSO

A1

3

(c) $e^{2x} = 1 + 2x + \frac{(2x)^2}{2} + \frac{(2x)^3}{6} + \dots$

$$= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$$

B1

1

(d) $1 - \cos x = \frac{1}{2}x^2 + \{o(x^4)\}$

B1

$$\frac{e^x (1+2x)^{\frac{1}{2}} - e^{2x}}{1 - \cos x} =$$

$$\frac{1+2x+x^2+\frac{2}{3}x^3-\left[1+2x+2x^2+\frac{4}{3}x^3\right]}{\frac{1}{2}x^2+\{o(x^4)\}}$$

Series used

M1

$$\lim_{x \rightarrow 0} \dots = \lim_{x \rightarrow 0} \frac{-x^2 + \{o(x^3)\}}{\frac{1}{2}x^2 + \{o(x^4)\}} =$$

A1F

$$\lim_{x \rightarrow 0} \frac{-1+o(x)}{\frac{1}{2}+o(x^2)} = -2$$

ft a slip but must see the intermediate stage

A1F

4

[16]

EXAM PAPERS PRACTICE

Q17.

(a) (i) $f(x) = \ln(1 + e^x)$:

$$f(0) = \ln 2$$

B1

$$f'(x) = \frac{e^x}{1+e^x} \quad f'(0) = \frac{1}{2}$$

Chain rule

M1

A1

$$f''(x) = \frac{(1+e^x)e^x - e^x e^x}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}$$

Quotient rule OE

M1

A1

$$f''(0) = \frac{1}{4}$$

so first three terms are:

$$f(x) = \ln 2 + \frac{1}{2}x + \frac{1}{4} \frac{x^2}{2!} = \ln 2 + \frac{1}{2}x + \frac{1}{8}x^2$$

CSO; AG

A1

6

(ii)
$$f'''(x) = \frac{(1+e^x)^2 e^x - e^x [2(1+e^x)e^x]}{(1+e^x)^4}$$

Chain rule with quotient/product rule

ft on $f'(x) = ke^x (1+e^x)^n$ (integer $n < 0$)

M1
A1ft

$$f'''(0) = \frac{4-4}{2^4} = 0$$

CSO; AG;

A1

{so coefficient of x^3 is zero}

All previous differentiation correct

3

SC for those not using Maclaurin's theorem: **maximum** of 4/9

(b)
$$\frac{1}{2}x + \frac{1}{8}x^2$$

B1

1

(c)
$$\ln\left(1 - \frac{x}{2}\right) = \left(-\frac{x}{2}\right) - \frac{1}{2}\left(-\frac{x}{2}\right)^2 + \frac{1}{3}\left(-\frac{x}{2}\right)^3 - \dots$$

B1

1

(d)
$$\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1 - \frac{x}{2}\right) = -\frac{x^3}{24} + \dots$$

Uses previous expansions to obtain first non-zero term of the form kx^3

M1

$$x - \sin x \approx x - \left[x - \frac{x^3}{3!} + \dots \right] \approx \frac{x^3}{3!} + \dots$$

B1

$$\left[\frac{\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1-\frac{x}{2}\right)}{x - \sin x} \right] = \frac{-\frac{1}{24}x^3 + \dots}{\frac{1}{6}x^3 + o(x^5)}$$

M1

$$= \frac{-\frac{1}{24}x^3 + \dots}{x^3 \left[\frac{1}{6} + o(x^2) \right]} = \frac{-\frac{1}{24} + \dots}{\frac{1}{6} + o(x^2)}$$

$$\lim_{x \rightarrow 0} \dots = -\frac{1}{4}$$

CSO

A1

4

[15]

Q18.

(a) $\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 \dots$

B1

1

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(b) (i) $f(x) = e^{\sin x} \Rightarrow f(0) = 1$

B1

$$f'(x) = \cos x e^{\sin x} \Rightarrow f'(0) = 1$$

M1A1

$$f'(x) = -\sin x e^{\sin x} + \cos^2 x e^{\sin x}$$

Product rule used

$$f'(0) = 1$$

M1A1

$$\text{Maclaurin } f(x) = f(0) + xf'(0) + \frac{x^2}{2} f''(0)$$

$$\text{so 1st three terms are } 1 + x + \frac{1}{2}x^2$$

CSO AG

A1

6

$$(ii) \quad f'''(x) = \cos x (\cos^2 x - \sin x) e^{\sin x} + \{2\cos x(-\sin x) - \cos x\} e^{\sin x}$$

M1A1

$f'''(0) = 0$ so the coefficient of x^3 in the series is zero

CSO AG

SC for (b): Use of series expansions.....max of 4/9

A1

3

(c) $\sin x \approx x.$

Ignore higher power terms in $\sin x$ expansion

B1

$$\frac{e^{\sin x} - 1 + \ln(1-x)}{x^2 \sin x} = \frac{-\frac{1}{3}x^3 + o(x^4)}{x^3}$$

Series from (a) & (b) used

M1

Numerator kx^3 (+...)

A1

$$= \frac{-\frac{1}{3} + o(x)}{1 + o(x^2)}$$

Condone if this step is missing

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1 + \ln(1-x)}{x^2 \sin x} = -\frac{1}{3}$$

On candidate's x^3 coefficient in (a) provided lower powers cancel

A1 ✓

4

[14]

Extra notes:

$$e^{\sin x} = 1 + \left(x - \frac{x^3}{3!} \dots\right) + \frac{1}{2!} \left(x - \frac{x^3}{3!} \dots\right)^2 + \frac{1}{3!} \left(x - \frac{x^3}{3!} \dots\right)^3 \dots$$

M1 for 1st 3 terms ignoring any higher powers than those shown.

A1 for all 4 terms (could be treated separately) ie last term often only comes into (b)(ii)

$$= 1 + x - \frac{x^3}{6} + \frac{1}{2}(x^2 - \dots) + \frac{1}{6}(x^3 - \dots) = 1 + x + \frac{1}{2}x^2$$

A1 (be convinced. ignore any powers of x above power 2)

Coefficient of x^3 : $-\frac{x^3}{6} + \frac{1}{6}x^3 = 0$

A1 (be convinced....ignore any powers of x above power 3)

Quite often the 2nd A mark is awarded before the 1st A1

Q19.

(a) (i) $(1+y)^{-1} = 1 - y + y^2 - \dots$

B1

1

(ii) $\sec x \approx \frac{1}{1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots}$

B1

$$= \left[1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots \right]^{-1} =$$

M1

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$$\left\{ 1 - \left(-\frac{x^2}{2} + \frac{x^4}{24} \right) + \left(-\frac{x^2}{2} + \frac{x^4}{24} \right)^2 \right\}$$

M1

$$= \left\{ 1 + \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^4}{4} + \dots \right\}$$

$$= 1 + \frac{x^2}{2} + \frac{5x^4}{24}$$

AG be convinced

A1;A1

5

Alternative: Those using Maclaurin

$$f(x) = \sec x$$

$$f(0) = 1; \underline{f'(x) = \sec x \tan x}; \{f'(0) = 0\}$$

(B1)

$$f''(x) = \sec x \tan^2 x + \sec^3 x; f''(0) = 1$$

Product rule oe

(M1)

$$f'''(x) = \sec x \tan^3 x + 5 \tan x \sec^3 x;$$

Chain rule with product rule OE

(m1)

$$f^{(iv)}(x) = \sec x \tan^4 x + 18 \tan^2 x \sec^3 x \dots$$

$$+ 5 \sec^5 x \Rightarrow f^{(iv)}(0) = 5$$

$\sec x \approx$ printed result

CSO AG

(A2)

(b) $f(x) = \tan x;$

$$f(0) = 0; f'(x) = \sec^2 x; \{f'(0) = 1\}$$

B1

$$f''(x) = 2 \sec x (\sec x \tan x); f''(0) = 0$$

$$f'''(x) = 4 \sec x \tan x (\sec x \tan x) + 2 \sec^4 x$$

Chain rule with product rule oe

M1

$$f'''(0) = 2$$

$$\tan x = 0 + 1x + 0x^2 + \frac{2}{3!}x^3 \dots = x + \frac{1}{3}x^3$$

CSO AG

A1

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3

Alternative: Those using otherwise

$$\dots = \frac{\sin x}{\cos x} \approx \left(x - \frac{x^3}{6} \dots\right) \left(1 + \frac{x^2}{2} \dots\right)$$

(M1) (A1)

$$= x + \frac{x^3}{2} - \frac{x^3}{6} \dots = x + \frac{1}{3}x^3 \dots$$

(A1)

(c)
$$\left(\frac{x \tan 2x}{\sec x - 1}\right) = \frac{x(2x + o(x^3))}{\frac{x^2}{2} + o(x^4)}$$

$$\tan 2x = 2x + \frac{1}{3}(2x)^3$$

B1

Condone $o(x^k)$ missing

M1

$$\frac{2+o(x^2)}{\frac{1}{2}+o(x^2)}$$

$$= 2$$

M1

$$\lim_{x \rightarrow 0} \left(\frac{x \tan 2x}{\sec x - 1} \right) = 4$$

ft on $2k$ after B0 for $\tan 2x = kx + \dots$

A1ft

4

[13]

EXAM PAPERS PRACTICE

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Examiner reports

Q1.

Almost all students were able to write down the correct expansion of e^{3x} . In part (b), the great majority of students used the 'hence' approach in which the most common error was to obtain a wrong coefficient of the x^2 term, normally $\frac{15}{8}$ but more surprisingly, $\frac{25}{2}$, in the binomial expansion. Only a small number of students used the alternative method of differentiation of $e^{3x} (1 + 2x)^{-1.5}$ and applying Maclaurin's theorem.

Q2.

In general, most students were able to obtain the printed result for the first derivative although a small number made errors on the way or failed to show sufficient steps. Both the direct differentiation and the use of a log law before differentiation methods were seen, although the second method resulted in fewer 'slips'. There were many good attempts seen to reach a correct expression for $\frac{d^4 y}{dx^4}$ in part (a)(ii) and most students showed that they knew how to find the relevant terms in the expansion by using their derivatives. The final mark for part (b) was only available to those who had correct derivatives in part (a). It was disappointing to see some less able students not attempting part (c), which should have been an easy mark to score with the aid of the formulae booklet. In general only the most able students were able to provide a convincing solution for part (d). It was not uncommon to see the expansions from parts (b) and (c) being written as a quotient of a logarithm rather than the log law being applied before the series expansions of the logarithms were substituted.

Q3.

This question, which tested differentiation and subsequent application of Maclaurin's theorem, proved to be a challenge to a significant number of students in part (b). Most students answered part (a) correctly with the majority using the chain rule and only a

minority finding and differentiating y^2 to reach $2y \frac{dy}{dx} = \cos x$. In part (b) the majority of students attempted to find the third derivative by differentiating $\frac{dy}{dx} = \frac{1}{2} (4 + \sin x)^{-1/2} (\cos x)$ twice. There were different forms presented with many mixtures of terms in x and y but a significant proportion of students gave incorrect expressions for the third derivative. Some

students obtained the correct expression for $\frac{d^2 y}{dx^2}$ but then did not attempt to differentiate the term $\frac{d^2 y}{dx^2} = \frac{1}{2} (4 + \sin x)^{-1/2} (-\sin x)$ when finding the value of $\frac{d^3 y}{dx^3}$ when $x = 0$,

presumably believing it would be zero since $\frac{1}{2} (4 + \sin x)^{-1/2} (-\sin x)$ itself is 0 when $x = 0$. A significant proportion of those students who used the implicit differentiation method made the same type of error as has been indicated in recent examiners' reports, namely,

differentiating $y \frac{dy}{dx}$ correctly as $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2$ but then writing this as $y \frac{d^2 y}{dx^2} + \frac{d^2 y}{dx^2}$ or, more costly, just writing the incorrect form. In part (c) many students scored the method mark but the final accuracy mark depended on correct earlier work and so was frequently not awarded.

Q4.

In part (a) candidates almost always used the chain rule correctly to find $\frac{dy}{dx}$ and frequently obtained a correct expression for $\frac{d^2y}{dx^2}$. A common error in finding the next two derivatives was to differentiate $\sec^2 2x$ incorrectly as $2\sec^2 2x \tan 2x$. This single error was not overly penalised with many such candidates losing just the one mark in part (a).

However, in part (b) it was disappointing to see so many candidates stating that the fourth derivative at $x = 0$ was -32 even though their expressions for $\frac{d^4y}{dx^4}$ clearly did not give rise to this value. Such candidates would have been better advised to look for their earlier error rather than make too much 'use' of the printed answer. Those who used the follow through value from their incorrect expression for $\frac{d^4y}{dx^4}$ generally scored 2 of the 3 marks for part (b), one more than those who stated -32 incorrectly.

In the final part of the question, in general only the more able candidates recognized and applied the 'Hence' by using ' $\ln(\sec^2 2x) = -2\ln(\cos 2x)$ '.

Q5.

This question on limits was generally answered well. Most students used the expansion of $\sin x$ from the formulae book and replaced x with $2x$ correctly to find the expansion of $\sin 2x$. In part (b) most students used the correct expansion of $\ln(1 + kx)$. The majority used their expansion from part (a) in the numerator, although some made an algebraic error in the subtraction of $\sin 2x$. Most then correctly divided numerator and denominator by x^3 before taking the limit. Some errors were seen in rearranging the equation $\frac{4}{3k} = 16$ to find the value of k . Not all students showed sufficient terms in the numerator and/or denominator for full marks to be scored.

Q6.

Most students used the chain rule correctly to obtain the correct answer for part (a) and then the quotient rule to find a correct expression for the second derivative, but not all of these students provided a convincing conversion to the required printed form. In part (c) a smaller proportion of students were completely successful. Students who used the expected method of keeping the right-hand side in terms of e , were able to find the third derivative but a significant minority did not then use the chain rule alongside the product rule to find the fourth derivative which frequently led to the incorrect answer

$\frac{d^4y}{dx^4} = -e^{-y} \left(\frac{dy}{dx} \right) - (e^{-y})^2$. It was more surprising to see students who had correctly

differentiated but then replaced the $-e^{-y} \left(\frac{dy}{dx} \right)^2$ term by $-e^{-y} \left(\frac{dy}{dx} \right)^2$ to reach

$-e^{-y} \left(\frac{d^2y}{dx^2} \right)$ to reach $\frac{d^4y}{dx^4} = 0$.

Students who found the third and fourth derivatives in terms of x gained credit but few could then write the fourth derivative in the required form. In part (d), most students displayed good knowledge of Maclaurin's theorem, but a significant number of students failed to score the final mark due to earlier errors.

Q7.

Most candidates were able to write down the required two expansions in parts (a).

In part (b)(i), candidates generally used the chain rule to find $\frac{dy}{dx}$ and then applied the product rule to find $\frac{d^2y}{dx^2}$. Although a common error was to differentiate $\sec^2x$ as

$2\sec x \tan x$, those who did find $\frac{d^2y}{dx^2}$ correctly often produced a convincing solution to reach the printed answer.

Some very good solutions were seen for parts (b)(ii) and (b)(iii) but it was disturbing to see some other candidates make no attempt to find the third derivative in part (b)(ii), yet claim that $f'''(0) = 3$ and write down the printed result in part (b)(iii). Clearly in such cases marks cannot be awarded.

Candidates who attempted part (c) generally showed a thorough understanding of the process required, including the need for explicitly reaching the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q8.

A majority of candidates were able to differentiate the given function correctly to find the first and second differentials although some made the second differentiation more complicated than was needed by first expressing their answer for the first derivative in terms of $\sin 2x$ and $\cos 2x$. Candidates should be aware that the expression for the derivative of $\sec x$ is in the formulae booklet.

In part (b), most candidates displayed good knowledge of Maclaurin's theorem but only those who had made no errors in earlier differentiations could score both marks for showing the printed result. It is pleasing to report that there was an improvement in candidates explicitly reaching the stage of a constant term with higher order terms in each of the numerator and denominator before taking the limit as $x \rightarrow 0$ in the final part of the question.

Q9.

Most candidates were able to find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ correctly, although some less able candidates failed to apply the chain rule. Although a few candidates in part (b) attempted to use the printed expansion of $\ln(1+x)$ from the formulae booklet instead of applying the prescribed method, the majority of candidates answered the question as instructed and showed good knowledge of Maclaurin's theorem. Many candidates failed to appreciate that in part (c) they had to replace x with $-x$, and instead multiplied both their x and x^2 terms by -1 . A significant minority of candidates in part (d) did not realise that they had to write the expression as the difference of their two expansions.

Q10.

This question on series expansions and the limiting process was generally answered very well, but a significant minority of candidates in part (b) did not explicitly reach the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q11.

Candidates' differentiation skills have improved significantly over recent series. Almost all candidates could correctly quote the series expansion for $\cos 4x$ although some failed to give their answer in its simplest form. The methods required to obtain the three derivatives were well understood and, although some algebraic errors were seen, there were many correct expressions presented.

Most candidates displayed good knowledge of Maclaurin's theorem but only those who had made no errors in earlier differentiations could score both marks for showing the printed result in part (b)(ii). There was a pleasing improvement in explicitly reaching the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$ in the final part of the question.

Q12.

Most candidates were able to write down the first four terms in the series expansion of e^{2x} , but in part (a)(ii) a significant minority of candidates ignored the request to use the binomial series expansion and instead resorted to differentiation of $f(x)$ and the use of Maclaurin's theorem. In part (b)(i), the most common error was to fail to use the chain rule

and thus to write the derivative of $\ln(1 + 2 \sin x)$ as $\frac{1}{1 + 2 \sin x}$.

Those candidates who correctly applied the chain rule usually went on to find the correct

expression for $\frac{d^2 y}{dx^2}$ by applying either the quotient rule or the product rule. Almost all

candidates who obtained an expression for $\frac{d^2 y}{dx^2}$ showed a correct understanding of Maclaurin's theorem in part (b)(ii).

Although the majority of candidates could correctly find the limiting value of the given expression as x tended to 0 in part (c), there were some incorrect solutions seen which just involved substituting $x = 0$ into the given expression.

Q13.

Most candidates were able to find $f'(x)$ correctly although some weaker candidates failed to apply the chain rule and were heavily penalised. The vast majority of the other candidates used the product rule (or quotient rule) but errors in differentiating $\sec^2 x$ were common. The majority of candidates showed good knowledge of Maclaurin's theorem but only those who had made no errors in earlier differentiations could score all 3 marks for showing the printed result in part (a)(i). Many weaker candidates failed to realise that the series expansion for $\sin 3x$ was required in part (b) and just stated the incorrect answer 0 for the limit. A significant minority of other candidates who used the expansion for $\sin 3x$ did not explicitly reach the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q14.

Although many candidates were able to answer part (a) correctly, it was clear, since factorials were present, that some had not realised that the series expansion for $\ln(1 + x)$ and its range of validity are both given in the formulae booklet. Methods of differentiation, chain rule and product/quotient rules, required in part (b) were either not known or not

applied correctly by a significant number of candidates. Some candidates left expressions unsimplified before carrying out further differentiation, often leading to expressions which required multiple applications of the chain, product and quotient rules. Such an approach clearly used up valuable examination time. Those who did simplify their expressions generally scored the available marks for parts (b). Although careless sign errors were seen in solutions to find the limit for part (c), the general understanding of the principles involved was better displayed than in some previous series.

Q15.

The examiners expected candidates to evaluate $3!$ in the expansion for $\sin 2x$, although credit was given retrospectively if the evaluation was left until part (c). Most candidates gave the correct expansion in part (a), and it was pleasing to find a greater proportion of the candidates than last summer applying the chain rule and product rule correctly in part (b)(i). Although Maclaurin's theorem was well known, a significant number of candidates who had obtained a wrong value for the second derivative in part (b)(i) tried to convince the examiners that it led to the printed result in part (b)(ii). Such candidates would have been better advised to look for their error in part (b)(i). Although many candidates scored the three marks in part (c), there were others who did not show the division of the numerator and denominator by x before finding the limit as x tended to zero.

Q16.

The vast majority of candidates answered both parts of part (a) correctly. Those who failed to use Maclaurin's theorem in part (a)(ii) were heavily penalised. Most candidates obtained the printed result in part (b), although others frequently failed to include all the terms in their working. Generally answers to part (c) were correct, although some solutions did not have the correct denominator for the x^3 term. In part (d), weaker candidates just replaced x by 0 and gave the wrong answer " $0/0 = 1$ ". In answering part (d), candidates were expected to show the division of terms by x^2 in the numerator and denominator so that the limiting value can be justifiably stated.

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Q17.

Many candidates, including those who had scored almost full marks on the previous five questions, failed to score many marks on this question. In part (a), a minority of candidates attempted to use series expansions instead of the prescribed Maclaurin's theorem. Although a special case was allowed, with a significant reduction in available marks, very few of these candidates scored any of them. For those using Maclaurin's theorem, most scored the first mark for finding $f(0)$ but a large number of candidates failed

to use the chain rule and differentiated $\ln(1 + e^x)$ incorrectly as $\frac{1}{1+e^x}$. Some of these candidates 'recovered' to score the next method mark but others just continued incorrectly to give subsequent derivatives in the form $(1 + e^x)^n$. Part (b) was not answered well, the

most common error being illustrated by

$$\ln\left(\frac{1+e^x}{2}\right) = \ln\frac{2}{2} + \frac{1}{2}\left(\frac{1}{2}x\right) + \frac{1}{2}\left(\frac{1}{8}x^2\right) = \frac{1}{4}x + \frac{1}{16}x^2$$

Most candidates gave the correct answer for part (c) and in part (d) a significant majority of candidates scored at least half the marks. Some candidates failed to use sufficient terms in the series expansion of $\sin x$.

Q18.

A common error in part (a) was to write the expansion of $\ln(1 - x)$ as the expansion of $-\ln(1 + x)$. It was disappointing to see so many candidates failing to use Maclaurin's theorem to answer part (b) especially as almost an identical question appears in the specimen paper for this unit. Those who did differentiate $f(x)$ generally obtained the first two derivatives correctly but then had difficulty when differentiating $\cos^2 x e^{\sin x}$. A significant minority of candidates failed to use the approximation for $\sin x$ in part (c).

Q19.

Average and weaker candidates scored relatively few marks in this question. The vast majority of candidates gained the mark for their answer to part (a)(i) but in part (ii) a common error for the lower grade candidate was to replace y by $\cos x$ and write ' $\sec x = 1 - \cos x + \cos^2 x \dots$ ' and then insert the expansion for $\cos x$. Those who used Maclaurin's theorem in part (a)(ii) were generally able to find correct expressions for $f(x)$ and $f'(x)$ but struggled to find $f'''(x)$ and $f^{(iv)}(x)$. Those who attempted part (b) generally gained the first mark for a correct start and first derivative but a significant minority omitted a '2' when finding $f'(x)$ as $\sec^2 x \tan x$. Those candidates who started by writing $\tan x = \sin x \sec x$ and applying the expansions for $\sin x$ and $\sec x$ generally obtained the printed answer convincingly. In part (c) the expansion for $\tan 2x$ was generally correct and most used the relevant expansions but then a significant number of candidates did not show the division of both the numerator and denominator by x^2 .